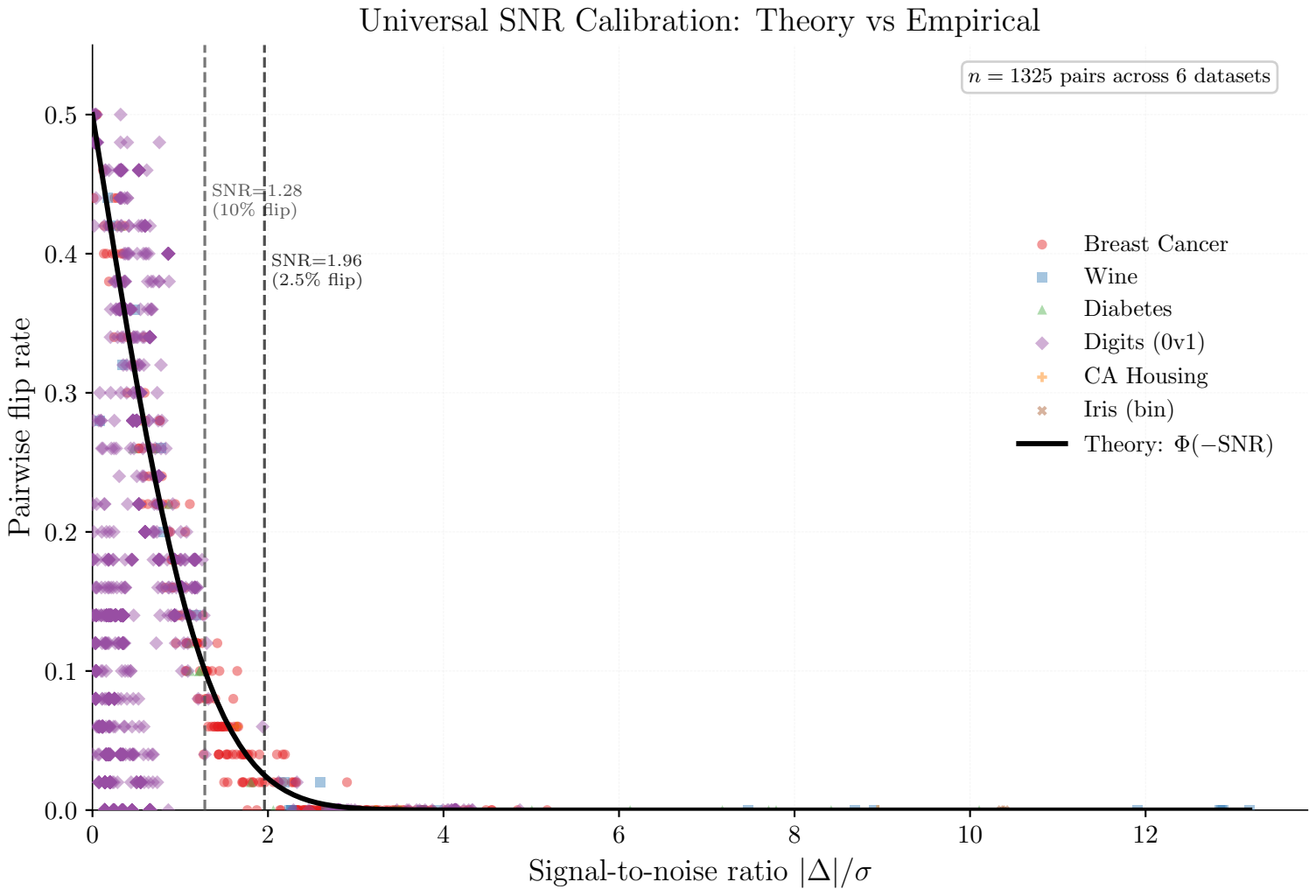
Universal SNR Calibration: Theory vs Empirical
$n = 1325$ pairs across 6 datasets
SNR=1.28
(10% flip)
SNR=1.96
(2.5% flip)
Breast Cancer
Wine
Diabetes
Digits (0v1)
CA Housing
Iris (bin)
Theory: $\Phi(-\text{SNR})$
Pairwise flip rate
Signal-to-noise ratio $|\Delta|/\sigma$
0.0
0.1
0.2
0.3
0.4
0.5
0
2
4
6
8
10
12